Test-Time Compositional Generalization in Diffusion Models via Prototype Discovery

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Abstract

Compositional generalization asks a model to produce what it has never seen by recombining what it knows. Diffusion methods for this problem typically assume that the reusable concepts are known in advance, provided through labels, prompts, or a learned concept library. We ask whether a pretrained diffusion model can instead discover its own reusable prototypes. A prototype is a mode of a noisy marginal $p_t(x_t)$. We perform gradient ascent on the learned score $s_\theta(x_t, t) \approx \nabla_{x_t} \log p_t(x_t)$ from a datum to recover a mode and fit a local Gaussian around it, and the noise level t sets the level of abstraction. On four pretrained models, up to a text-to-image model with 1.45B parameters, the prototypes recovered from different data are distinct at fine noise levels and become less separated as noise increases, while a datum's own prototype changes sharply across levels. A single datum therefore induces several distinct, reusable prototypes rather than one prototype under increasing blur. With these discovered prototypes, at test time we select those that explain a single out-of-distribution query and compose them into a product of experts whose closed-form score guides the composable diffusion sampler. On held-out compositions of ColorMNIST and CelebA, the samples generated from one query generalize to other instances of the unseen class, where the same sampler conditioned on trained concept tokens lands on neighboring classes. A pretrained diffusion model thus already contains prototypes at many levels of abstraction, and they can be recovered and composed without supervision.

1 Introduction

Human intelligence is compositional. After learning familiar parts, relations, or words, people can recombine them to understand and produce novel configurations. This ability has long motivated work in cognitive science and linguistics. In these fields, systematic generalization is often linked to reusable primitives and operators for recombination (Fodor & Pylyshyn, 1988; Lake et al., 2017). It is also a central challenge for deep learning, where neural networks may interpolate well within a training distribution but fail when familiar primitives appear in unfamiliar combinations (Lake & Baroni, 2018; Ruis et al., 2020). The key question is therefore not only whether models learn useful representations, but whether they can discover reusable units and compose them at test time given queries. Work in deep learning often addresses this problem by designing representations or architectures that support composition. Neural module networks, for example, assemble learned components from linguistic or program structure (Andreas et al., 2016b;a). Benchmarks such as SCAN test systematic recombination (Lake & Baroni, 2018; Ruis et al., 2020). Visual generation methods likewise ask whether learned objects, attributes, or concepts can be recombined out of distribution. Despite this progress, these approaches often assume that the relevant primitives are supplied by the benchmark, labels, prompts, or a predefined concept library.

A probabilistic view offers a complementary perspective. Concepts can be treated as high-probability regions in an energy landscape, while composition requires satisfying multiple constraints at once. This idea appears in Hopfield attractors and mixed states (Hopfield, 1982; Amit et al., 1985), product-of-experts models (PoE) (Hinton, 2002), and recent energy-based or diffusion composition methods (Du et al., 2020; Liu et al., 2022; Du et al., 2023). These methods typically start composition

with component distributions or conditioning signals that are already available. Experts can be learned during training, but discovering the prototypes needed for a new query remains a separate problem. In visual domains, the relevant primitive may depend on the query and emerge at different levels of abstraction, such as digit identity, color, texture, or facial attributes. A fuller account should therefore discover reusable prototypes from the model's learned density and compose them without fixed vocabularies or external labels.

Diffusion models offer a natural setting for this view. A pretrained diffusion model learns time-indexed scores $s_\theta(x_t, t) \approx \nabla_{x_t} \log p_t(x_t)$ for the noisy marginals $p_t(x_t)$. We use those scores for discovery rather than reverse sampling. Each noise level smooths the data distribution at a different scale. At each level, the local mode reached from a datum defines a prototype whose granularity is set by t. A single datum can therefore induce several prototypes across noise levels. Ascending the score from a datum recovers its mode at a given level, and a local Gaussian centered at that mode describes the prototype's extent.

We recover prototypes across all noise levels on four pretrained models, from a CIFAR-10 DDPM to a text-to-image diffusion model, and analyze their behavior. The prototypes of different data are far apart at fine noise levels and become less separated as noise increases, while a datum's own prototype changes sharply across levels. A single datum thus yields several distinct prototypes across scales rather than one prototype under increasing blur. We then ask whether an out-of-distribution datum can be written as a combination of prototypes the model already has. Our method selects prototypes for one such query and combines them into a product-of-experts distribution that drives the composable diffusion sampler (Liu et al., 2022; Ho & Salimans, 2022). On ColorMNIST and CelebA benchmarks of held-out compositions, the samples from a single query generalize to other instances of the unseen class, where the same sampler conditioned on trained concept tokens lands on neighboring classes.

Contributions.

- We recover prototypes from a pretrained diffusion model without labels, a concept library, or additional training. Each prototype is a mode of a noisy marginal $p_t(x_t)$, reached by score ascent and described by a local Gaussian, and the noise level t sets its level of abstraction.
- We show on four pretrained models, from CIFAR-10 and CelebA-HQ DDPMs to a text-to-image model with 1.45B parameters, that a single datum can produce distinct and diverse prototypes across noise levels. The prototypes of different data are far apart at fine noise levels and become less separated as noise increases, while a datum's own prototype changes sharply across levels.
- We turn discovery into test-time compositional generalization. Prototypes selected for a single OOD query form a product of experts whose closed-form score guides the composable diffusion sampler toward other instances of the query's unseen class. On ColorMNIST and CelebA benchmarks of held-out attribute compositions, the generated samples match a near copy of the query on attribute correctness while exceeding it in diversity, whereas the same sampler conditioned on trained concept tokens (Liu et al., 2022) lands on neighboring classes and query editing at a matched attribute correctness has a higher KID and a lower diversity.

2 Related work

Prototype discovery Prior work on prototype discovery aims to identify reusable abstractions between pixels and labels, such as parts, attributes, or object-level factors. It has studied such concepts as interpretable directions in representation space (Bau et al., 2017; Kim et al., 2018; Ghorbani et al., 2019), as discriminative prototypes (Chen et al., 2019), or as composable generative units (Du et al., 2020; Liu et al., 2023). These methods reveal meaningful visual structure, but they are often discriminative or rely on external supervision.

From a statistical perspective, prototypes can be interpreted as local density modes, with examples assigned to their basins of attraction. This connects prototype discovery to mode-seeking and hierarchical clustering methods such as mean-shift (Comaniciu & Meer, 2002), density-derivative-ratio mode estimation (Sasaki et al., 2018), cluster trees (Chaudhuri & Dasgupta, 2010), and hierarchical prototype models (Wang et al., 2025). We instead recover prototypes by seeking modes of a pretrained diffusion model at multiple noise levels. Ladder VAE and NVAE organize their latent variables into hierarchies (Sønderby et al., 2016; Vahdat & Kautz, 2020). Diffusion models

admit a similar latent-variable interpretation across noise levels (Sohl-Dickstein et al., 2015; Ho et al., 2020; Kingma et al., 2021). Recent work also shows that diffusion timesteps organize information from coarse semantic structure to fine detail, with phase-transition behavior at intermediate noise levels (Sclocchi et al., 2025; Huang et al., 2024). Motivated by this view, we discover prototypes by seeking modes of the smoothed marginal $p_t(x)$ using the pretrained score $\nabla_x \log p_t(x)$, with no additional training, external supervision, or manually specified taxonomy.

Compositional generalization Compositional generalization is the ability to recombine familiar parts into novel configurations, a central problem in cognitive science, linguistics, and machine learning (Fodor & Pylyshyn, 1988; Lake et al., 2017; Lake & Baroni, 2018). In AI, this ability has been studied through diagnostic benchmarks such as SCAN and gSCAN (Lake & Baroni, 2018; Ruis et al., 2020), modular architectures that combine learned neural components (Andreas et al., 2016b;a), and data-augmentation or semantic-parsing methods that encourage systematic recombination (Andreas, 2020; Yin et al., 2021). These approaches differ in implementation, but typically share the assumption that the relevant primitives are already specified by the benchmark, the language, or a predefined module set.

A complementary line of work provides a probabilistic framework for composition. Product-of-experts (PoE) models combine constraints by multiplying concept densities (Hinton, 2002). Energy-based models use this principle to compose visual concepts (Du et al., 2020). Composable diffusion models implement analogous composition by adding concept-specific scores (Liu et al., 2022), with later work improving sampling through MCMC corrections (Du et al., 2023). Classifier-free guidance provides a related score-combination mechanism for conditional generation (Ho & Salimans, 2022). The experts can be learned during training, but composition typically uses component distributions or conditioning signals that are already available.

Our work connects prototype discovery with compositional generalization, following the view that systematic generalization requires both reusable parts and an operator for recombining them (Fodor & Pylyshyn, 1988; Lake et al., 2017). The work closest to our setting, Liu et al. (2023), learns a shared library of concept embeddings from unlabeled images, whereas we discover query-specific concepts at test time as modes of the unconditional diffusion density and compose their local Gaussian approximations without a predefined concept vocabulary.

Diffusion models Diffusion models learn time-indexed score fields for the noisy marginals $p_t(x)$ and are typically used as generative samplers (Vincent, 2011; Song & Ermon, 2019; Ho et al., 2020; Song et al., 2021). We repurpose this learned score geometry for prototype discovery. Instead of immediately running the reverse sampler, we use $s_\theta(x, t) \approx \nabla_x \log p_t(x)$ to seek modes of intermediate marginals. Recent analyses support this view by showing that diffusion timesteps encode structure across scales. Class identity disappears sharply while lower-level features persist across a phase transition (Sclocchi et al., 2025). Generative trajectories commit to classes at sharp symmetry-breaking and speciation times (Raya & Ambrogioni, 2023; Biroli et al., 2024). At lower noise levels, Gaussian mixtures can approximate the learned score more accurately than a single Gaussian (Wang & Vastola, 2024). We measure how the modes of p_t change across noise levels and use them as prototypes at the granularity determined by t.

3 Prototype Discovery Across Diffusion Noise Levels

In this section, we describe how to discover prototypes from a pretrained diffusion model given a query image, with no labels, no concept library, and no additional training. We first show that the noisy marginals expose density modes at different scales (Section 3.1), then describe how to recover those modes through score ascent (Section 3.2). Each recovered mode is a prototype whose granularity is set by the noise level. Section 4 analyzes the modes found from pretrained models, and Section 5 uses them for test-time compositional generalization.

3.1 Noisy marginals expose density modes across scales

We work in the discrete-time diffusion setting (Sohl-Dickstein et al., 2015; Ho et al., 2020). Let $x_0 \in \mathbb{R}^d$ denote a noise-free datum drawn from p_{data}, and let

$$x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1-\bar{\alpha}_t}\, \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, I), \quad t \in \{1, \dots, T\}, \tag{1}$$

be the forward-diffused state under schedule $\{\bar{\alpha}_t\}$. The noisy marginal is

$$p_t(x_t) = \int p_{\text{data}}(x_0)\, \mathcal{N}\Big(x_t; \sqrt{\bar{\alpha}_t}\, x_0,\, (1-\bar{\alpha}_t) I\Big)\, dx_0, \tag{2}$$

whose score $\nabla_{x_t} \log p_t(x_t)$ the network approximates through the noise-prediction head $\varepsilon_\theta(x_t, t)$ (Song & Ermon, 2019; Song et al., 2021),

$$s_\theta(x_t, t) = -\varepsilon_\theta(x_t, t) \big/ \sqrt{1-\bar{\alpha}_t}. \tag{3}$$

Eq. (2) states that, up to the rescaling by $\sqrt{\bar{\alpha}_t}$, the marginal p_t is the data distribution smoothed by a Gaussian kernel of bandwidth $\sqrt{1-\bar{\alpha}_t}$. The noise schedule therefore defines a scale-space over p_{data} (Witkin, 1983). Assume that p_{data} is locally Gaussian around each datum. Then p_{data} can be represented as a Gaussian mixture, $p_{\text{data}}(x_0) = \sum_k \pi_k\, \mathcal{N}(x_0; \mu_k, \Sigma_k)$ for some unknown mixing factor π_k. This representation leaves p_{data} as intractable as before but gives the smoothed marginal in closed form,

$$p_t(x_t) \;=\; \sum_k \pi_k\, \mathcal{N}\Big(x_t;\; \sqrt{\bar{\alpha}_t}\, \mu_k,\; \bar{\alpha}_t \Sigma_k + (1-\bar{\alpha}_t) I\Big). \tag{4}$$

Every component retains its rescaled center and gains variance $(1-\bar{\alpha}_t)I$. As t grows, the rescaled components overlap more, and nearby density peaks can become less distinct or merge. This motivates recovering prototypes at multiple smoothing scales. Mean-shift locates density modes at a fixed bandwidth (Comaniciu & Meer, 2002; Sasaki et al., 2018), while cluster trees track the connected components of density superlevel sets as the density threshold varies (Chaudhuri & Dasgupta, 2010). The diffusion forward process fixes the bandwidth ladder, with each level corresponding to a latent state (Kingma et al., 2021). The pretrained score estimates the log-density gradient at each level through Eq. (3).

Several lines of work motivate studying prototypes across noise levels. Sclocchi et al. (2025) identify a phase transition in the denoising process at which class identity is lost while low-level features persist, and attribute it to the hierarchical structure of natural data. Related analyses find sharp symmetry-breaking and speciation times at which generative trajectories commit to a class (Raya & Ambrogioni, 2023; Biroli et al., 2024). At lower noise levels, Gaussian mixtures can approximate the learned score more accurately than a single Gaussian (Wang & Vastola, 2024). Recent work interprets the modes of noisy diffusion marginals as a hierarchy of concept prototypes (Wang et al., 2026). This view also parallels categorization in cognitive science, where concepts form hierarchies whose intermediate levels carry the most predictive structure (Rosch et al., 1976). Hierarchical prototype models impose this structure through explicit taxonomic architectures (Wang et al., 2025), whereas flat prototype models learn parts without any hierarchy (Chen et al., 2019). We recover modes of p_t at several noise levels and treat them as prototypes at the granularity set by t, without imposing an explicit taxonomy.

3.2 Prototype discovery via mode ascent

By a *mode* of p_t we mean a local maximum of the noisy marginal, i.e., a solution of

$$x_t^* \in \operatorname*{arg\,max}_{x_t \in U}\; p_t(x_t) \quad \text{for an open neighborhood } U \text{ of } x_t^*. \tag{5}$$

Solving Equation (5) in closed form is intractable because p_t is available only implicitly through its score. We therefore locate modes by gradient ascent on the log-density,

$$x_t^{(i+1)} = x_t^{(i)} + \eta\, s_\theta\Big(x_t^{(i)},\, t\Big), \tag{6}$$

initialized from a noised copy of the query datum,

$$x_t^{(0)} = \sqrt{\bar{\alpha}_t}\, x_q + \sqrt{1-\bar{\alpha}_t}\, \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, I).$$

Using the unconditional score ensures that ascent explores the full marginal p_t rather than the conditional manifold of any trained class. We set the step to $\eta = 1 - \bar{\alpha}_t$ and halve it whenever an iterate jumps across the mode, so no step size is tuned (Section F.1).

Modes yield clean prototypes. A random draw from p_t is visibly noisy by Eq. (1), yet a mode of p_t is noise-free. This follows from Tweedie's formula (Robbins, 1956; Miyasawa, 1961; Efron, 2011; Vincent, 2011), which expresses the posterior mean of the clean datum as

$$\hat{x}_0(x_t) := \mathbb{E}[x_0 \mid x_t] = \frac{1}{\sqrt{\bar{\alpha}_t}}\Big(x_t + (1-\bar{\alpha}_t)\,\nabla_{x_t} \log p_t(x_t)\Big). \tag{7}$$

At a mode, the score vanishes, so Eq. (7) reduces to $\hat{x}_0(x_t^*) = x_t^*/\sqrt{\bar{\alpha}_t}$. The added Gaussian noise therefore has zero conditional mean. We treat the recovered modes as noise-free prototypes in this sense, even though posterior averaging may blur image detail.

At sufficiently low noise, Gaussian smoothing approximately preserves a well-separated local mode m of p_{data}. For the corresponding mode x_t^* of p_t, this gives

$$x_t^* \approx \sqrt{\bar{\alpha}_t}\, m, \qquad \hat{x}_0(x_t^*) \approx m \in \operatorname{supp}(p_{\text{data}}). \tag{8}$$

Following the mixture form of Eq. (4), we approximate the local neighborhood of p_t around a recovered mode with a Gaussian $q_j = \mathcal{N}(m_j, \Sigma_j)$, where m_j is the recovered mode and Σ_j is a diagonal Laplace estimate computed at the mode using four Hutchinson probes (Hutchinson, 1990). The pair (m_j, Σ_j) is the discovered prototype at level t_j, with m_j its location and Σ_j its extent.

Modes found at multiple noise levels translate into several prototypes from one datum. Section 4 measures the separation between prototypes from different data and the changes in a single datum's prototype across noise levels. Section A shows an example on Fashion-MNIST.

4 Empirical Analysis of Prototypes Across Noise Levels

We examine how the recovered prototypes vary across data and noise levels. We measure the separation between prototypes from different data at each level and track how a single datum's prototype changes across levels. We use four pretrained models, without labels: the released unconditional DDPMs for CIFAR-10 and CelebA-HQ (Ho et al., 2020), DiT-XL/2 (Peebles & Xie, 2023) trained on ImageNet (0.68B parameters), and the text-to-image latent diffusion model (LDM) of Rombach et al. (2022) trained on LAION-400M (1.45B). For a fixed set of held-out images, or of the model's own samples on the two larger models, we consider noise levels between $t = 10$ and 990. At each level, we add noise to the image, ascend the score as in Eq. (6), and fit the local Gaussian of Section 3.2 at the recovered mode. Section B gives the full configuration.

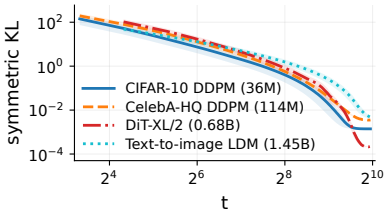


Figure 1: Symmetric KL between the local Gaussians of two data at the same level on a log scale. Each datum is compared with eight others.

Metrics. We use two measurements, both computed directly from the recovered Gaussians.

- *Prototype separation (symmetric KL).* To quantify how far apart the prototypes of different data are, we compute the symmetric KL divergence between the local Gaussians recovered from two data at the same level. Each held-out datum is paired with the prototypes of other data, and the divergence is aggregated over pairs, seeds, and data at every level.
- *Normalized displacement.* To quantify how a datum's prototype departs from its initial image, we measure the distance between them after normalizing both locations to unit norm, $d_i^{(t)} = \left\| m_i^{(t)}/\|m_i^{(t)}\| - m_i^{(0)}/\|m_i^{(0)}\| \right\|$, where $m_i^{(t)}$ is the mode recovered from datum i at level t and $m_i^{(0)}$ is its initial image. Normalizing removes the amplitude decay $\sqrt{\bar{\alpha}_t}$ of the forward process, so the displacement changes only when the prototype itself moves. A sudden rise marks a phase change, an interval over which the prototype changes far faster than in the neighboring levels.

Prototypes become less separated as noise increases. Figure 1 shows the separation between prototypes recovered from different data. At fine levels the local Gaussians are well separated. As t grows, the separation falls by four to six orders of magnitude on all four models. Prototypes from different data thus become less distinguishable by this measure at higher noise levels. This trend is expected as the forward process approaches a single Gaussian. We interpret this loss of separation as coarsening, consistent with the scale-space view of structure at different smoothing levels (Witkin, 1983).

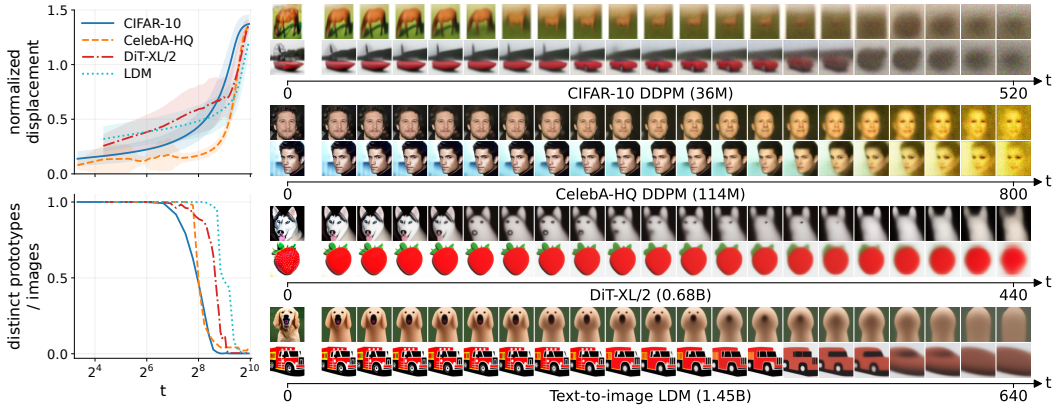


Figure 2: The left panel shows the normalized displacement between each recovered prototype $m_i^{(t)}$ and the initial image $m_i^{(0)}$ (top) and the number of distinct prototypes per image (bottom). The right panel shows two examples per model, each row an image followed by its prototypes at increasing t.

Different levels yield distinct prototypes. Figure 2 tracks the normalized displacement of a datum's prototype from its initial image and the number of distinct prototypes, alongside examples of prototypes recovered at increasing t. The displacement is nonuniform across noise levels. On all four models it changes slowly over the fine levels and then rises sharply, from $t \approx 450$ on CIFAR-10, $t \approx 500$ on CelebA-HQ, and $t \approx 600$ on the two larger models. The region of rapid change marks a phase change, where the prototype reorganizes far faster than in the surrounding levels, consistent with the phase transitions and speciation times reported for diffusion models (Sclocchi et al., 2025; Raya & Ambrogioni, 2023; Biroli et al., 2024). Every datum has its own prototype at fine levels, and the data come to share prototypes as t grows, until all images of a model share one or two. The prototype images show what is changing. On CelebA-HQ, facial detail and identity persist through the fine levels and give way to more generic faces across the rise. CIFAR-10 loses object identity at a finer level, matching its earlier rise. These changes show that one datum can yield distinct prototypes at different noise levels, with changes in visual structure beyond a gradual loss of detail. For additional analyses on which images share a prototype first and on what a prototype keeps of combinations the model has and has not seen, we refer to Section B.

These observations motivate collecting prototypes at several noise levels for a given query. We next test whether the prototypes recovered from one out-of-distribution query support generalization to other instances of its unseen class.

5 Test-Time Compositional Generalization from Discovered Prototypes

Section 4 shows that one datum can yield distinct prototypes across noise levels. We now test whether composing their local Gaussian densities supports generalization from a single OOD query. The target is a held-out combination of familiar attributes. Successful samples must preserve this combination while varying beyond the particular image supplied as the query.

5.1 From discovered prototypes to a composable score

Given one out-of-distribution query x_q, we run mode ascent at several noise levels (Section 3.2) from noised copies of x_q and of the model's own samples, and collect a pool of candidate prototypes $\{q_j\}$. A greedy submodular selection keeps $K = 8$ prototypes that jointly best explain the query per dimension, and the selected experts are combined into a weighted product of experts (Hinton, 2002). Each expert is first rescaled to the clean variable x_0 so that all experts share the random variable required by the product (Section C). The product q_{T} remains Gaussian with a closed-form mean and covariance, and the score of its forward-diffused marginal is analytic at every t.

Sampling then reduces to composable diffusion (Liu et al., 2022). Here the backbone is class-conditional, with noise prediction $\varepsilon_\theta(x_t, t, c)$ for trained classes $c \in C$ and a null token $\emptyset$ for the unconditional branch. The sampler keeps the unconditional prediction of the backbone (Ho

& Salimans, 2022) and adds to it the analytic score contribution of the PoE of the discovered prototypes, where composable diffusion adds that of a trained concept token (Eq. (15)). The comparison therefore focuses on the source of the conditional experts. We refer to Section C for an overview (Figure 10) and full details of the selection objective, the closed-form PoE, and the sampler.

5.2 Experimental setup

Each benchmark separates attribute combinations into seen classes for training and held-out classes for evaluation. Every held-out class recombines familiar attribute values. At test time, the generator receives one image from that class and must produce other instances with the same defining attributes. The target label and other members of the class are used only for evaluation.

ColorMNIST. A 32×32 RGB rendering of MNIST with three primitive factors, digit identity (10 values), digit color (4 values), and background color (4 values). The Cartesian product yields 160 compositional slots, of which 120 are seen by the backbone during training and 40 are held out as OOD classes. To form the held-out split, we exclude four pairs of digit and background colors across all ten digits. All individual digit and color values remain represented in training.

CelebA (compositional). A benchmark built from binary CelebA (Liu et al., 2015) face attributes such as hair color, smiling, beard, and gender. Each unique realized combination of 14 binary attributes defines a compositional class. Of the 74 realized combinations, 51 form the seen split and 23 are held out. Images are rendered at 128×128. Each held-out combination differs from at least one seen combination by a single attribute, testing generalization to compositions near familiar classes.

The class definition leaves room for variation within each target. ColorMNIST samples can differ in handwriting while retaining the same digit identity and colors. CelebA samples can differ in identity, pose, and illumination while sharing the selected attributes. These variations are the intended outputs of generalization. A near copy of the query can retain all attributes without producing a new instance, while a visually plausible face or digit can still belong to the wrong compositional class.

Both backbones are class-conditional DDPMs trained with classifier-free guidance dropout on the seen split only and kept fixed at test time. Prototype discovery uses the unconditional branch of the same backbone whose conditional branches are used by Comp. 1 and Comp. 3. Appendix D details the datasets and Appendix E the backbone training.

5.3 Hypothesis, baselines, and evaluation

Hypothesis. We hypothesize that prototypes recovered at different noise levels provide complementary information about the query's held-out attribute combination. Their product should therefore guide the pretrained backbone toward samples that preserve the target attributes while varying beyond the query. We evaluate both requirements together, since copying the input can preserve its attributes without generating other instances, while unconstrained variation can leave the target class.

Baselines. For each OOD class, we evaluate on 100 queries (10 on ColorMNIST). Each query is processed separately, with no pooling of images from the same class. **Ours** is the composable-diffusion sampler driven by the analytic PoE score of the discovered prototypes. Sampling settings for our method and every baseline are listed in Section F.2. We compare with three baselines.

- **Composable diffusion (Comp. 1, Comp. 3).** Both instantiate the sampler of Liu et al. (2022) with trained concept tokens from the same backbone. They test whether selecting or composing familiar class distributions is sufficient for the query's unseen combination. For each query, every trained class $c \in C$ is scored by the average DDPM loss of x_q under that conditional,

$$\ell(c) := \mathbb{E}_{t \in \mathcal{T}_{\text{eval}},\, \varepsilon} \left\| \varepsilon - \varepsilon_\theta(\sqrt{\bar{\alpha}_t} x_q + \sqrt{1-\bar{\alpha}_t}\varepsilon,\, t,\, c) \right\|^2,$$

and the k smallest-loss classes are kept. Comp. 1 conditions the sampler on the single best trained token. Comp. 3 composes the three best through the multi-concept combination of the same sampler with weights $w_k = \mathrm{softmax}_k(-\ell(c_{(k)})/\tau_{\mathrm{tk}})$.

- **SDEdit.** The baseline of Meng et al. (2022) forward-noises the query to an intermediate level t_0 and denoises it with the unconditional model. It uses the query as a whole image without explicitly discovering or composing prototypes. This tests whether direct image editing can preserve the target attributes while producing other instances of the class. We sweep t_0 from 50 to 500 (Section H).

Evaluation protocol and metrics. Every method turns one query into a batch of 16 samples, each generated with 50 DDIM steps. Each batch is scored against the held-out samples of the query's class, excluding the query itself, and scores are averaged over queries and then over classes. This measures *generalization*, whether the batch inferred from one image covers the class. We use three measurements.

- **KID.** The kernel inception distance (Bińkowski et al., 2018) between each batch and the held-out members measures generation quality. We refer to Section G for additional discussions.
- **Diversity.** A batch can vary widely yet leave the queried class, so plain spread is not the right measure here. We use the quality–diversity decomposition of determinantal point processes (Kulesza & Taskar, 2012) to balance proximity to the query with diversity within the batch. A batch $\{x_i\}_{i=1}^N$ scores

$$\tfrac{1}{N}\log\det(QSQ) \;=\; \tfrac{2}{N}\sum\nolimits_i \log q_i \;+\; \tfrac{1}{N}\log\det S,$$

  where q_i measures how relevant sample x_i is to the query, Q collects the q_i on its diagonal, and S measures the pairwise similarity within the batch. The first term rewards proximity to the query and the second rewards diversity within the batch. We assess agreement with the class's defining attributes using attribute correctness, defined below. Section G gives the kernel definitions and evaluation details.
- **Attribute correctness.** To measure composition, we report the fraction of the class's defining attributes that a sample carries. On ColorMNIST, attribute values are read from the nearest composition mean in pixel space. On CelebA, each attribute is judged by whether the sample's CLIP embedding lies closer to the positive or negative attribute mean. Averaging the judgments over the class's defining attributes $\mathcal{A}$ gives

$$\mathrm{Corr}(x) \;=\; \tfrac{1}{|\mathcal{A}|}\sum\nolimits_{a\in\mathcal{A}} \mathbf{1}\big[\hat{y}_a(x) = y_a\big],$$

  where $\hat{y}_a(x)$ is the judgment and y_a the class's label for attribute a. We refer to Section G for the embeddings and additional details.

5.4 Results and analysis

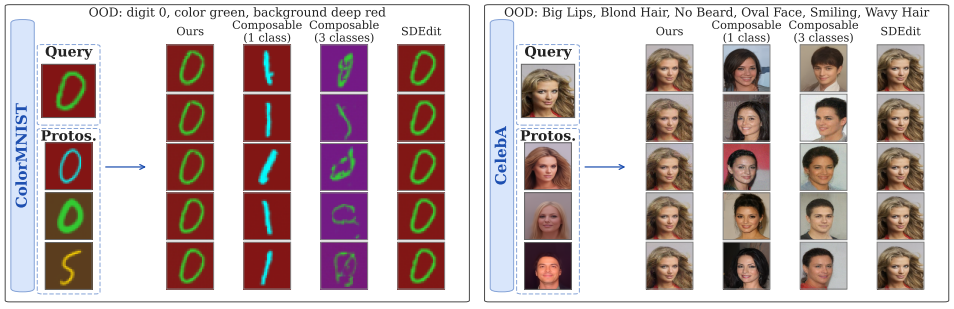


Figure 3: Examples of prototypes discovered from OOD queries representing unseen compositions (three of the $K = 8$ are shown). Appendix K provides additional qualitative results.

Our method matches SDEdit on attribute correctness and improves on it everywhere else (Table 1). SDEdit at $t_0 = 50$ returns near copies of the query, so its attribute correctness is essentially that of a real member, 92.9% correct on ColorMNIST and 78.3% correct on CelebA. Our generated samples reach 92.5% and 77.0% correct, within about a point of the copies, while improving

Table 1: Test-time composition from a single OOD query (mean$_{\pm SE}$). Best in **bold**, second underlined.

	ColorMNIST			CelebA		
Method	Attr. Corr. ↑	KID ↓	Diversity ↑	Attr. Corr. ↑	KID ↓	Diversity ↑
Comp. 1	$37.7_{\pm 1.2}$	$0.166_{\pm 0.008}$	$-26.1_{\pm 4.6}$	$63.4_{\pm 1.2}$	$\mathbf{0.082}_{\pm 0.003}$	$-5.24_{\pm 0.10}$
Comp. 3	$39.3_{\pm 1.1}$	$0.130_{\pm 0.004}$	$-25.2_{\pm 4.3}$	$62.2_{\pm 1.0}$	$\underline{0.092}_{\pm 0.003}$	$-5.05_{\pm 0.09}$
SDEdit (t_0=50)	$\mathbf{92.9}_{\pm 0.9}$	$0.107_{\pm 0.005}$	$-3.40_{\pm 0.06}$	$\mathbf{78.3}_{\pm 2.1}$	$0.218_{\pm 0.006}$	$-4.21_{\pm 0.04}$
SDEdit (t_0=200)	$92.3_{\pm 0.9}$	$\underline{0.082}_{\pm 0.004}$	$\underline{-2.17}_{\pm 0.05}$	$\underline{77.1}_{\pm 2.1}$	$0.179_{\pm 0.005}$	$\underline{-3.21}_{\pm 0.03}$
Ours	$\underline{92.5}_{\pm 0.9}$	$\mathbf{0.079}_{\pm 0.004}$	$\mathbf{-2.12}_{\pm 0.04}$	$77.0_{\pm 2.0}$	$0.166_{\pm 0.004}$	$\mathbf{-3.13}_{\pm 0.02}$

KID (0.079 versus 0.107 on ColorMNIST, 0.166 versus 0.218 on CelebA) and scoring the highest diversity of all methods on both datasets. The diversity gap is expected from the construction of the score, because near copies stay relevant to the query yet are nearly identical to each other, and identical samples earn little diversity. A larger t_0 lowers the KID of SDEdit only together with its attribute correctness, on CelebA from 0.218 at 78.3% to 0.131 at 73.7% (Table 10). KID is therefore comparable only at equal attribute correctness. At $t_0 = 200$, where SDEdit matches our attribute correctness, our KID is lower and our diversity higher on both datasets, and our KID is lower at every attribute correctness that both methods reach (Figure 11). Matching a copy on attribute correctness while being more diverse and closer in KID to the held-out samples indicates that the discovered prototypes carry information about the held-out class distribution rather than the query alone. The composable diffusion baselines (Comp. 1 and Comp. 3) fail on both attribute correctness and diversity. Their generated samples vary the most in raw pixel distance, yet their diversity is the lowest of all methods, because the variation falls outside the queried class and the relevance to the query collapses (Figure 3). We argue that the failure comes from retrieval. Retrieval can only return trained classes, and the query's held-out class is excluded from that set by construction. On average over queries, the three classes retrieved by Comp. 3 jointly cover all defining attribute values in only 4.3% of ColorMNIST queries and 18.1% of CelebA queries. Sampling from these classes therefore caps attribute correctness near chance. Comp. 1 and Comp. 3 score 37.7% and 39.3% on ColorMNIST, where assigning attribute values at random gives 20%. On CelebA they have the lowest KID, but at 63.4% and 62.2% attribute correctness against our 77.0% (Section G). These baselines share our sampler and differ only in where the concepts come from, so the gap is attributable to concept discovery. We refer to Section G for additional discussion where the attribute correctness threshold is calibrated for attribute imbalance.

Together, we show that a single OOD sample suffices for the discovered prototypes to induce a distribution over its unseen class. The resulting sampler keeps the attribute correctness of a near copy while improving on its KID and diversity, which neither the composable diffusion baselines nor the edited query achieves. Section I varies the number of selected prototypes K.

6 Conclusion, Limitations, and Future Work

We studied test-time compositional generalization by recovering prototypes from a pretrained diffusion model. A prototype is a mode of a noisy marginal $p_t(x_t)$, reached by ascending the learned score from a datum and described locally by a Gaussian, with the noise level setting its level of abstraction. On four pretrained models of different size and type, the prototypes of different data are distinct at fine noise levels and become less separated as noise increases, while a datum's own prototype changes sharply across levels. A single datum can therefore provide several distinct prototypes at different scales. Composing the prototypes discovered from one out-of-distribution query into a product of experts lets the composable diffusion sampler generate other instances of the query's held-out class. On ColorMNIST and CelebA, the generated samples match a near copy of the query on attribute correctness while exceeding it in diversity, whereas the same sampler conditioned on trained concept tokens lands on neighboring classes. These findings suggest that diffusion models encode reusable prototypes beyond point reconstruction or class-level retrieval. The same reading of the score field may extend to other domains where diffusion models are used, including language modeling (Li et al., 2022; Nie et al., 2025) and robotic control (Chi et al., 2023).

Limitations. Our protocol follows the standard compositional generalization setting, where primitive factors are known and test cases are unseen recombinations. We evaluate the composition

of density modes as concept prototypes through generalization to unseen attribute combinations, but do not directly analyze compositional structure in the model's internal neural representations. The setting remains relevant for foundation models, which acquire rich primitives during pretraining, so many novel instances are closer to unseen compositions than to unseen concepts. Human-level intelligence, however, also learns new primitives and reuses them in later compositions (Fodor & Pylyshyn, 1988; Lake et al., 2017; 2015). Our framework sometimes discovers novel-looking primitives by interpolating between known ones (Appendix L). A full mechanism for detecting, consolidating, and reusing genuinely new primitives remains future work.

AI use statement

In this work, we used generative AI tools for codebase organization and tracking experimental results. We have not used generative AI tools for generating the core research ideas or conducting autonomous research, and tasks involving formal proofs are not applicable to this work. Additionally, we used generative AI tools for manuscript writing and proofreading. We have reviewed all AI-assisted work. The authors checked the AI-generated text and code. We take responsibility for the final content of this work, including text, claims, or artifacts produced with the aid of generative AI.

Ethics statement

We conduct experiments on publicly available image benchmarks using diffusion models, without recruiting human participants or collecting new personal data. The face datasets contain images of real people, so applications should consider privacy, biases in the data and attribute annotations, and the potential misuse of generated faces. Licenses for existing assets are summarized in Appendix N.

Reproducibility statement

Sections 3 and 5 describe prototype discovery, composition, and evaluation. Appendices B and C detail the prototype analysis, prototype selection, product-of-experts construction, and sampling. Appendices D, E, and F specify the datasets and held-out splits, model architectures, training configurations, and hyperparameters. Evaluation metrics and computational resources are documented in Appendices G and J, and the SDEdit sweep and the ablations in Appendices H and I.

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A Modes of Diffusion Marginals Across Noise Levels



(a) $t = 0$ (b) $t = 50$ (c) $t = 200$ (d) $t = 300$ (e) $t = 500$

Figure 4: Modes of the noisy marginals $p_t(x_t)$ learned by a DDPM on Fashion-MNIST (Xiao et al., 2017). The $t = 0$ panel shows clean reference images, while larger t panels show prototypes recovered by mode ascent at progressively noisier marginals. As t increases, fine instance details are smoothed away and the recovered prototypes show coarser object structure. The noise levels therefore order the recovered prototypes by scale.

To motivate our use of mode finding for prototype discovery, we visualize local modes of the noisy marginals $p_t(x_t)$ learned by a DDPM trained on Fashion-MNIST (Xiao et al., 2017). Using the mode-ascent procedure in Section 3.2, we recover prototypes at several noise levels. As t increases, the marginals become progressively smoother. The recovered modes lose fine instance details and show coarser object structure. They provide prototypes at different levels of abstraction.

B Additional analysis

On CIFAR-10 and CelebA-HQ, the analysis of Section 4 uses the released unconditional DDPM checkpoints (Ho et al., 2020) without any fine-tuning, and Section B.1 describes the two larger models. For each dataset we take a fixed set of held-out images, 512 for CIFAR-10 and 48 for CelebA-HQ, and the grid of noise levels $t \in \{10, 20, \ldots, 990\}$. At every (image, level) pair, we forward-noise the image using Eq. (1) and run the ascent in Eq. (6) with the fixed step $\eta = (1 - \bar{\alpha}_t)/2$. The ascent stops at the first step at which the root mean square (RMS) of the predicted noise over the d dimensions, $\|\varepsilon_\theta(x_t, t)\|/\sqrt{d}$, is below 0.10, with a cap of 200 steps. By Eq. (3) the predicted noise is the rescaled score, so the score has nearly vanished where the ascent stops, and we take this point as the recovered mode. The local covariance is a diagonal Laplace estimate at the recovered mode, computed using four Hutchinson probes with finite-difference step 5×10^{-3}. We repeat the full procedure for five ascent seeds. All curves and prototype images use the recovered mode, so every divergence pairs the mode with the covariance measured at it.

The separation of Figure 1 is computed between the local Gaussians of two data at the same level. The five ascent seeds are first pooled into one Gaussian per datum and level by the law of total variance. Each datum i is then compared with the eight fixed partners $i{+}1, \ldots, i{+}8$, taken cyclically over the held-out set, the per-coordinate symmetric KL is reduced to its median over coordinates, and the eight values are averaged. The curves show the mean over the held-out data with a band of ± 1 standard deviation. Normalized displacement is the Euclidean distance between the recovered mode $m_i^{(t)}$ and the initial image $m_i^{(0)}$, with both normalized to unit norm, averaged over seeds and images. Prototype images are shown from a single ascent seed.

Number of distinct prototypes. For the number of distinct prototypes we run the ascent for 200 steps from one noised copy of every image, at $t \in \{20, 60, \ldots, 980\}$ (on CIFAR-10 also at $t = 10$, 30, 40, and 50), and group the recovered modes. A mode joins the first group whose first member lies within an RMS distance of $0.05\sqrt{\bar{\alpha}_t}\, L$, where L is the median RMS norm of the clean images, and opens a new group otherwise. The number of groups is the number of distinct prototypes at that level, and two images share a prototype when their modes lie within this distance of each other.

B.1 Larger models

DiT-XL/2 (Peebles & Xie, 2023) is a class-conditional transformer with 0.68B parameters, trained on ImageNet (Deng et al., 2009) at 256×256 in the latent space of an autoencoder. The text-to-image latent diffusion model (LDM) of Rombach et al. (2022) has 1.45B parameters, a U-Net with 0.87B and a text encoder with 0.58B, and is trained on LAION-400M (Schuhmann et al., 2021). We use the public checkpoints of both models.

The images are 256 samples of each model itself. For DiT-XL/2 they are 16 samples for each of 16 ImageNet classes in four groups, dogs (golden retriever, German shepherd, Siberian husky, pug), birds (goldfinch, robin, hummingbird, flamingo), vehicles (convertible, fire engine, school bus, sports car), and fruit (Granny Smith, strawberry, orange, banana). For the LDM they are 16 samples for each of 16 prompts of the form "a photo of a golden retriever", with the same objects and a green apple in place of the Granny Smith. Both sets are sampled with 50 DDIM steps and classifier-free guidance, at scale 4 for DiT-XL/2 and 6 for the LDM.

Mode ascent runs in latent space with the unconditional branch of each model, the null class for DiT-XL/2 and the empty prompt for the LDM, with the fixed step $\eta = 1 - \bar{\alpha}_t$ and $\eta = (1 - \bar{\alpha}_t)/2$, respectively. The measurements of Section 4 are computed as above in latent space, with one ascent seed, at $t \in \{20, 40, \ldots, 980\}$ for DiT-XL/2 and $t \in \{20, 60, \ldots, 980\}$ for the LDM. For the LDM at $t \geq 460$, the ascent for the number of distinct prototypes continues until the RMS of the predicted noise is below 10^{-3}, for at most 1000 steps.

Table 2: Number of distinct prototypes that the images of each model reach at selected noise levels t.

Model	Images	$t = 100$	220	300	380	460	580	660	740	860
CIFAR-10 DDPM	512	508	321	139	14	1	1	1	1	1
CelebA-HQ DDPM	48	48	47	9	4	2	2	2	2	1
DiT-XL/2	256	255	237	222	166	37	1	1	1	1
LDM	256	256	256	256	248	140	112	21	6	1

Table 2 shows that the images of every model come to share prototypes as the noise level grows. Each image has its own prototype at fine levels, and all images of a model share one or two prototypes at coarse levels, from $t = 460$ on the two DDPMs, $t = 560$ on DiT-XL/2, and $t = 820$ on the LDM. Over the grid, the separation of Figure 1 falls from 104 to 2.2×10^{-4} on DiT-XL/2 and from 48.6 to 4.9×10^{-3} on the LDM.

Figure 5 shows four examples per model in the display of Figure 2. On the two larger models, as on the two DDPMs, the prototype of an image changes from level to level and shows a simpler version of its object at a higher noise level. The examples are images whose prototypes change visibly from level to level and still show an object at high noise levels.

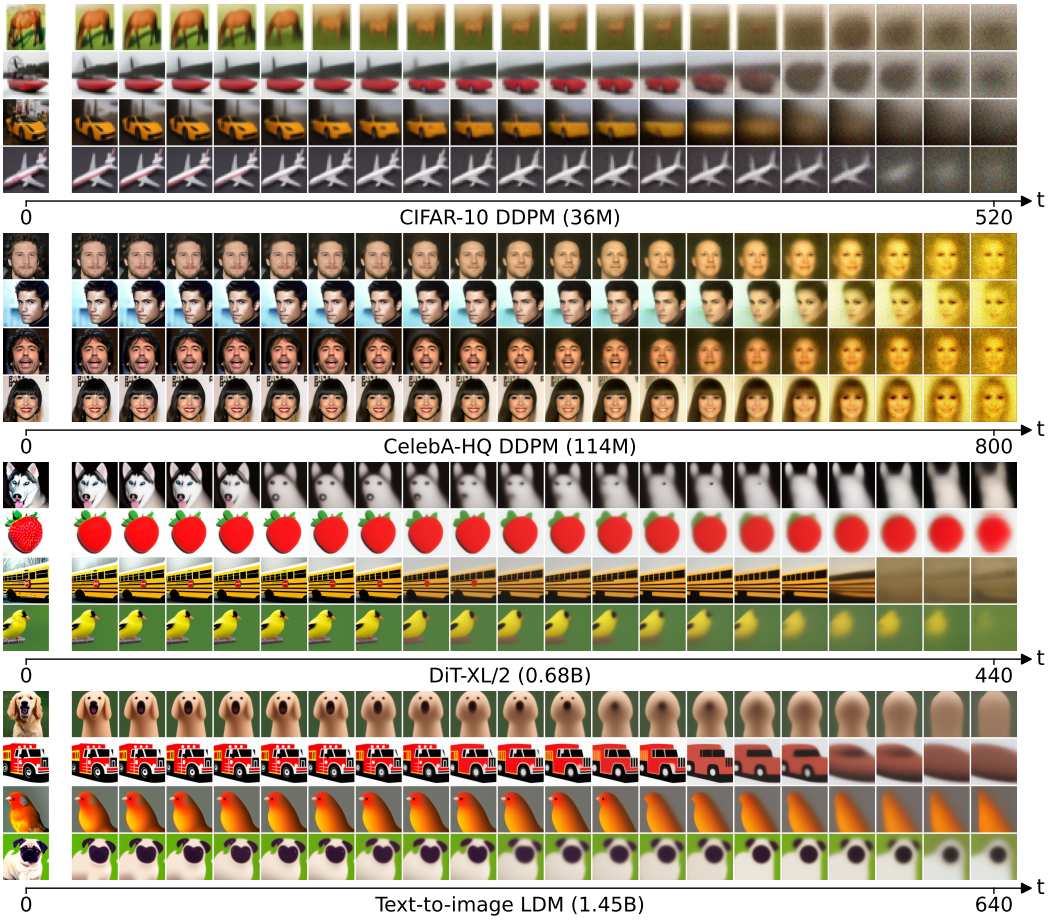


Figure 5: Prototypes at increasing noise levels, four examples per model. Each row shows the original image, a held-out image for the two DDPMs and a sample of the model for the two larger models, followed by its prototype at 20 increasing levels t. The ascent at level t stops at an RMS of the predicted noise below 0.10, and the recovered mode is decoded with Eq. (7) and, for the two latent models, with the autoencoder of the model. The rows of DiT-XL/2 show a Siberian husky, a strawberry, a school bus, and a goldfinch, and those of the LDM a golden retriever, a fire engine, a robin, and a pug.

B.2 Images with the same content share a prototype first

We ask which images come to share a prototype first as the noise level grows. On the two larger models, a pair of images comes from the same class or prompt, from the same group only, or from different groups.

Figure 6 shows that pairs with the same content share a prototype first on both models. On DiT-XL/2 at $t = 440$, 44% of the pairs from the same class share a prototype, against 21% of the pairs from different groups. On the LDM at $t = 580$, 8.7% of the pairs from the same prompt share a prototype, against 4.2% of the pairs from different groups. The prototypes at coarse levels therefore group images by their content.

A shared prototype could be the center of the images that share it. To test this, for every set of at least two images that share a prototype, we measure the RMS distance between the prototype, rescaled to the clean variable, and the mean of the images, in units of the spread of the set, the RMS distance of its images from their mean. The prototype lies 1.2 to 1.8 spreads from the mean on DiT-XL/2 and 1.0 to 1.3 spreads on the LDM, at least as far as the images themselves, so a shared prototype is not the center of its images.

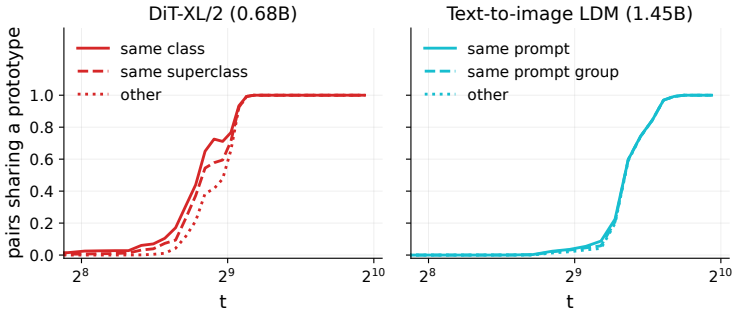


Figure 6: Share of image pairs that share a prototype against t, by pair type, for DiT-XL/2 and the LDM.

B.3 A prototype keeps what the model has learned

A prototype could be the datum under a growing amount of blur. It would then depend only on the datum and on the noise level, and not on what the model has learned. We test this on the ColorMNIST backbone of Section 5, which is trained on 12 of the 16 pairs of digit color and background color (Section D). We take 40 OOD queries, whose color pair is held out, and recolor each digit to two seen pairs, one that keeps the background and one that keeps the digit color. From each image we run 32 ascents of 40 steps at every noise level and read the color pair of each prototype with the judge of Section G.

Table 3: Share of ascents (%) whose prototype keeps the color pair of the image they start from, on the ColorMNIST backbone (40 queries, 32 ascents each).

Image the ascent starts from	$t = 100$	$t = 200$	$t = 300$	$t = 400$
Held-out color pair	60.0	22.3	9.7	9.8
Seen color pair, same background	100.0	92.5	89.8	85.7
Seen color pair, same digit color	94.8	90.0	92.5	96.8

Table 3 shows that a prototype keeps a color pair that the model has seen and replaces one that it has not. At $t = 300$, 89.8% of the ascents from a seen pair with the same background keep their pair, against 9.7% of the ascents from a held-out pair, which end on a seen pair instead. The images differ only in the color pair, so what a prototype keeps is set by what the model has learned and not by the amount of smoothing. Figure 7 shows these shares against the noise level and against the number of ascent steps, and Figure 8 shows single ascents step by step. A draw from p_t is visibly noisy, the first steps remove the noise, and the later steps move a held-out pair to a seen one while a seen pair stays.

Usual and unusual colors on the LDM. The LDM has no held-out combination, but some combinations are rarer than others, such as a blue banana. We ask whether a prototype keeps a usual color longer than an unusual one. We generate six objects, each in its usual color and in an unusual one, a banana (yellow, blue), a strawberry (red, blue), an orange (orange, purple), a fire engine (red, green), a school bus (yellow, pink), and a flamingo (pink, green). For every prompt of the form "a photo of a blue banana" we generate 32 images and keep up to 16 that CLIP ViT-B/32 (Radford et al., 2021) assigns to their own prompt rather than to the other prompt of the pair, which leaves 96 images in a usual color and 91 in an unusual one. We run mode ascent with the unconditional branch at $t \in \{100, 200, \ldots, 600\}$, decode the iterate after 3, 10, 40, and 150 ascent steps, and ask CLIP again which of the two prompts it matches, where chance is 50%.

Figure 9 shows that the ascent keeps the usual color longer than the unusual one. At $t = 400$ after 10 ascent steps, the usual color is kept in 95% of the images and the unusual one in 81%. The LDM thus behaves as the ColorMNIST backbone does.

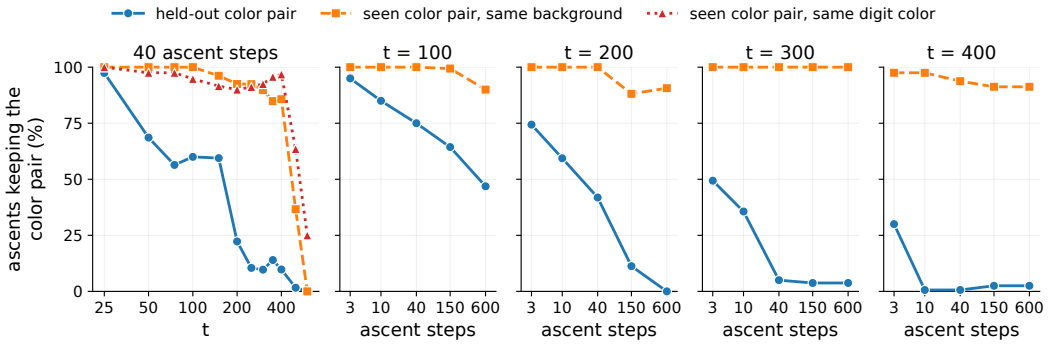


Figure 7: Share of ascents on the ColorMNIST backbone whose prototype keeps the color pair of the image they start from, for held-out color pairs and for the same digits recolored to seen pairs. The left panel shows the share after 40 ascent steps against the noise level t. The right panels show the share against the number of ascent steps at four noise levels.

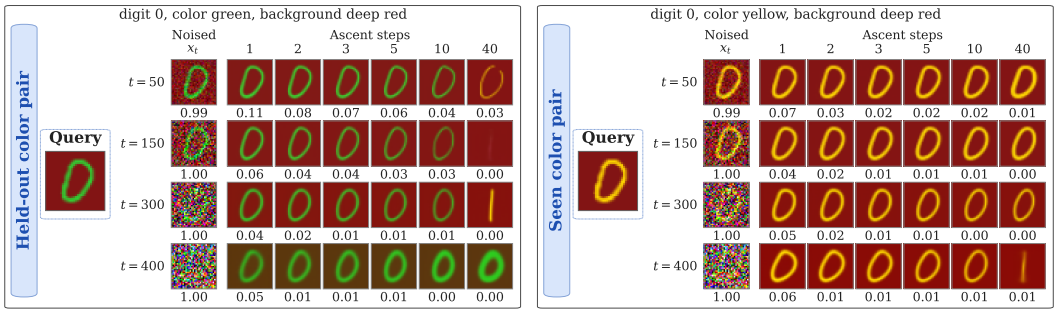


Figure 8: Mode ascent step by step on the ColorMNIST backbone, from a noised copy of a held-out query (digit 0, green on deep red, left) and of the same digit recolored to a seen pair with the same background (yellow on deep red, right), at $t = 50$, 150, 300, and 400 (rows). Each row shows the noised copy x_t and the result after 1, 2, 3, 5, 10, and 40 ascent steps, decoded with Eq. (7), with the RMS of the predicted noise under each image.

C Product-of-experts composition and sampling

This appendix provides full details of the method in Section 5.1. Figure 10 shows the procedure on one held-out ColorMNIST query.

Candidate pool. At each noise level of the grid in Table 7, mode ascent starts from noised copies of the query and from noised copies of the model's own unconditional samples, which are drawn once and shared by all queries. Each mode recovered at level t is rescaled by $\sqrt{\bar{\alpha}_{50}/\bar{\alpha}_t}$ and ascended again at $t = 50$, so that a prototype found at a high noise level is refined into a mode of a low-noise marginal, and the local Gaussian is fitted there.

The local Gaussians of Section 3.2, discovered for the query x_q, are rescaled to the clean variable as $q_j(x_0) = \mathcal{N}(x_0; m_j, \Sigma_j)$ with $m_j = x_t^{*,j}/\sqrt{\bar{\alpha}_{t_j}}$, so that every expert shares the random variable x_0. In what follows, greedy selection keeps K of them, indexed by $S_K \subseteq \{1, \dots, M\}$.

Weighted product of experts. Each candidate is an expert over the shared variable x_0, so selected prototypes combine through a weighted product of experts (Hinton, 2002). We use per-dimension weights $w_j \in [0, 1]^d$ under the diagonal Gaussian approximation. The product remains Gaussian, $q_T = \mathcal{N}(\mu_T, \Sigma_T)$, with closed-form parameters

$$q_T(x_0) \propto \prod_{j \in S_K} q_j(x_0)^{w_j}, \quad \Sigma_T^{-1} = \sum_{j \in S_K} \mathrm{Diag}(w_j)\, \Sigma_j^{-1}, \quad \mu_T = \Sigma_T \sum_{j \in S_K} \mathrm{Diag}(w_j)\, \Sigma_j^{-1} m_j, \tag{9}$$

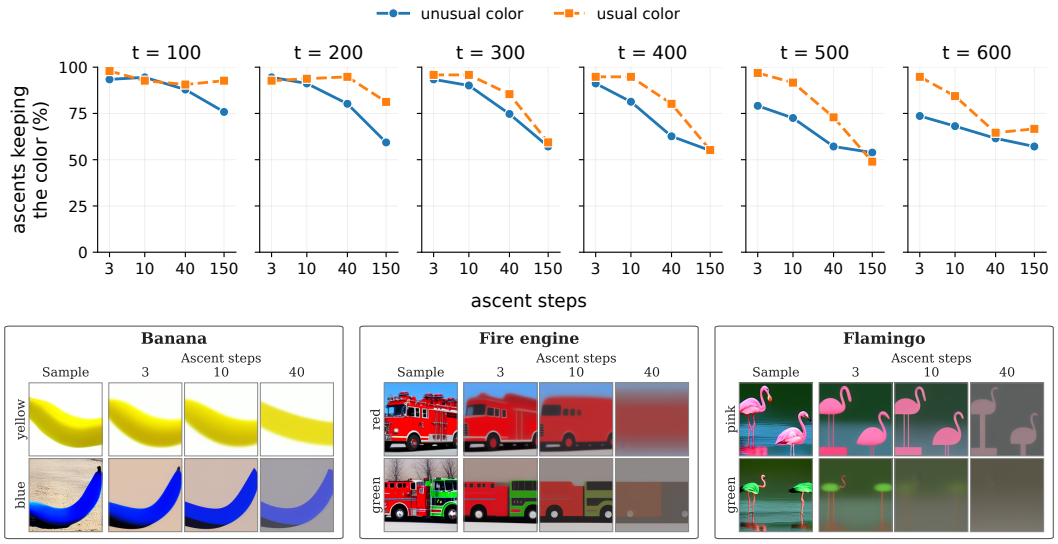


Figure 9: Usual and unusual colors on the LDM. Top: share of images whose decoded iterate CLIP still assigns to the image's own color prompt rather than to the other prompt of its pair (chance 50%), against the number of ascent steps at six noise levels, for 96 images of six objects in their usual color and 91 in an unusual color. Bottom: three objects in their usual color (upper row of each block) and in an unusual color (lower row), with the sample of the model and the result after 3, 10, and 40 ascent steps at $t = 300$.

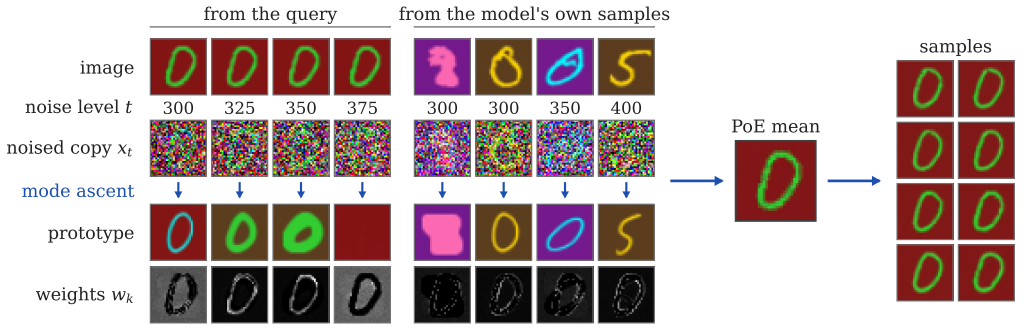


Figure 10: Test-time composition for one held-out ColorMNIST query (digit 0, green on deep red). Each column is one of the $K = 8$ prototypes that greedy selection keeps from the candidate pool. From top to bottom, a column shows the image the ascent starts from, which is the query or one of the model's own unconditional samples, its noised copy x_t, the prototype that mode ascent reaches from it, and the per-dimension weights of this prototype in the product of experts (white is 1). The PoE, whose mean is shown, guides the sampler toward other instances of the held-out class.

where exponentiation is dimension-wise. In each dimension, a prototype contributes to the composed mean according to its query-dependent weight and local precision. Different prototypes can therefore dominate different dimensions of the target distribution used for compositional sampling (Du & Mordatch, 2019; Du et al., 2020; Liu et al., 2022; Du et al., 2023).

Greedy submodular selection. Write

$$\ell_{j,r}(x_q) := \log \mathcal{N}\left(x_{q,r};\, m_{j,r},\, \sigma^2_{j,r}\right), \qquad r \in \{1, \ldots, d\},$$

where $\Sigma_j = \mathrm{Diag}(\sigma^2_{j,1}, \ldots, \sigma^2_{j,d})$. Here, $\ell_{j,r}(x_q)$ is the contribution of prototype j to the log-likelihood of x_q at dimension r. Under the diagonal PoE of Eq. (9), the log-likelihood decomposes per

dimension, so for nonempty S we use the coverage surrogate

$$F(S) \;:=\; \sum_{r=1}^{d} \max_{j\in S} \ell_{j,r}(x_q), \tag{10}$$

with $F(\varnothing) := \sum_{r=1}^{d} \min_{1\le j\le M} \ell_{j,r}(x_q)$ for the empty set. The function F is monotone and submodular (Nemhauser et al., 1978; Krause & Golovin, 2014). We initialize with the best singleton, $S_1 = \{\arg\max_j F(\{j\})\}$, and grow the set by the largest marginal gain,

$$j_{i+1} \;=\; \operatorname*{arg\,max}_{j\,\in\,\{1,\dots,M\}\setminus S_i} \Delta_j(S_i), \qquad \Delta_j(S_i) \;:=\; F(S_i\cup\{j\}) - F(S_i), \qquad S_{i+1} = S_i \cup \{j_{i+1}\}, \tag{11}$$

for $i = 1, \dots, K-1$. After selection, the per-dimension weight of each selected prototype is the temperature-controlled softmax of its per-dimension log-likelihood over the selected set,

$$w_j(r) \;=\; \frac{\exp\big(\ell_{j,r}(x_q)/\tau\big)}{\sum_{j'\in S_K}\exp\big(\ell_{j',r}(x_q)/\tau\big)}, \qquad j \in S_K, \tag{12}$$

so prototypes with higher likelihoods receive larger weights in that dimension, and similar likelihoods give similar weights.

Analytic score and compositional sampling. Because q_{T} is Gaussian in clean data space, so is its forward-diffused counterpart $q_{\mathrm{T}}^{(t)}(x_t) = \mathcal{N}\big(x_t; \sqrt{\bar\alpha_t}\mu_{\mathrm{T}},\ \bar\alpha_t\Sigma_{\mathrm{T}} + (1-\bar\alpha_t)I\big)$, whose score is

$$\nabla_{x_t} \log q_{\mathrm{T}}^{(t)}(x_t) = -\big[\bar\alpha_t\Sigma_{\mathrm{T}} + (1-\bar\alpha_t)I\big]^{-1}\big(x_t - \sqrt{\bar\alpha_t}\,\mu_{\mathrm{T}}\big). \tag{13}$$

Composable diffusion combines concepts by adding conditional score contributions to the unconditional score, each normally a learned conditional score (Liu et al., 2022). Our discovered composition is not a trained token. Its contribution is Eq. (13) in closed form. Converting to the noise-prediction parameterization via Eq. (3) gives

$$\varepsilon_{q_{\mathrm{T}}}(x_t, t) = \sqrt{1-\bar\alpha_t}\,\big[\bar\alpha_t\Sigma_{\mathrm{T}} + (1-\bar\alpha_t)I\big]^{-1}\big(x_t - \sqrt{\bar\alpha_t}\,\mu_{\mathrm{T}}\big). \tag{14}$$

We add this contribution to the unconditional prediction of the backbone (Ho & Salimans, 2022) and evaluate it at the denoised state $\tilde x_t = x_t - \sqrt{1-\bar\alpha_t}\,\varepsilon_\theta(x_t, t, \emptyset)$, so the combined prediction becomes

$$\hat\varepsilon(x_t, t; c_*) = \varepsilon_\theta(x_t, t, \emptyset) + \varepsilon_{q_{\mathrm{T}}}(\tilde x_t, t). \tag{15}$$

Eq. (15) samples the discovered composition by adding the analytic PoE prediction to the null prediction, without a trained conditional token for c_*. Written for the clean variable, with $\hat x_0^\emptyset = \tilde x_t/\sqrt{\bar\alpha_t}$ the estimate of the unconditional branch from Eq. (7), Eq. (15) reads

$$\hat x_0 \;=\; \hat x_0^\emptyset + \kappa_t\big(\mu_{\mathrm{T}} - \hat x_0^\emptyset\big), \qquad \kappa_t \;=\; (1-\bar\alpha_t)\,\big[\bar\alpha_t\Sigma_{\mathrm{T}} + (1-\bar\alpha_t)I\big]^{-1}, \tag{16}$$

which is the mean of the product of two Gaussians over x_0, the estimate of the backbone $\mathcal{N}\big(\hat x_0^\emptyset, \frac{1-\bar\alpha_t}{\bar\alpha_t}I\big)$ and the PoE q_{T}. The pull toward μ_{T} is strong at large t, where the estimate of the backbone is uncertain, and vanishes as $t \to 0$, so Eq. (15) needs no guidance weight.

Scale of the guidance. Two settings determine the pull of the PoE. First, Σ_{T} is multiplied by a scale S in Eq. (16). A larger S widens the PoE and lowers KID together with attribute correctness (Section H). Second, the pull $\kappa_t(\mu_{\mathrm{T}} - \hat x_0^\emptyset)$ is averaged over $N \times N$ pixel blocks before it is applied, so the PoE sets the content at that scale, such as colors and layout, and the backbone draws the detail below it.

Several PoEs per query. On CelebA the candidate pool is split into four parts. Each part gives its own PoE of K prototypes by the selection above, and the 16 samples are divided equally among the four PoEs, so a query yields four compositions (Figure 13). On ColorMNIST one PoE and four PoEs give the same result (92.5% and 92.4% attribute correctness), and we use one.

Table 4: ColorMNIST primitive attributes and split.

Attribute	Cardinality
Digit identity	10
Digit color	4
Background color	4
Total compositions	160
Seen slots (training)	120
Unseen slots (OOD)	40
Per-slot images	1,000
Image format	32×32 RGB

D Additional benchmark details

The ColorMNIST primitive values are digit colors in {yellow, green, cyan, pink} and background colors in {deep red, navy, dark purple, dark brown}, a muted high-contrast palette chosen so that no digit color collapses against a background under low-light rendering. The 40 unseen slots are formed by holding out four (digit color, background color) pairs, namely (yellow, dark purple), (green, deep red), (cyan, dark brown), and (pink, navy), across all ten digits, giving $4 \times 10 = 40$ OOD slots. The remaining 120 slots train the backbone.

Table 5: CelebA compositional benchmark (Liu et al., 2015). A compositional class is a unique combination of 14 binary CelebA attributes (Black/Blond/Brown/Gray hair, Wavy hair, No_Beard, Smiling, Big_Nose, Big_Lips, Oval_Face, Male, Sideburns, Wearing_Necklace, Bald). The 74 realized combinations used in this experiment are partitioned into a seen tier used for backbone training and a holdout tier accessible only at test time.

Attribute / split	Value
Number of binary attributes	14
Realized compositional classes	74
Seen classes (training)	51
train images	26,775
in-distribution validation images	3,825
Holdout classes (OOD)	23
images	9,914
Image format	128×128 RGB

The 23 unseen CelebA classes are chosen so that their attribute combination differs from at least one seen class by a single attribute, providing a controlled OOD setting where each held-out class is reachable from a near neighbor in the seen set. Class ids are remapped contiguously to $\{0, \ldots, 119\}$ for ColorMNIST and $\{0, \ldots, 50\}$ for CelebA before being passed to the backbone's class embedding. The backbone never sees an image from the held-out tier during training.

E Backbone architecture and pretraining

Each backbone is a class-conditional `diffusers` `UNet2DModel` (Ho et al., 2020) trained with classifier-free guidance dropout (Ho & Salimans, 2022). We use `google/ddpm-cifar10-32` for the 32×32 ColorMNIST backbone and `google/ddpm-celebahq-256` (with `sample_size` overridden to 128) for the 128×128 CelebA backbone. The class-embedding table is sized to $|\mathcal{C}| + 1$ to reserve one row for the null token used during CFG dropout. Training uses the DDPM scheduler with a linear β schedule, $T = 1000$ training timesteps, and ε-prediction. ColorMNIST is trained on a single GPU in full precision. CelebA is trained on 8 GPUs in bf16 mixed precision with an EMA on UNet weights, and we evaluate the EMA checkpoint at training step 500,000. Table 6 lists the per-dataset training hyperparameters. Architectural fields not listed (block out-channels, layers-per-block, attention resolutions) follow the base UNet configuration for each dataset.

Table 6: Backbone training hyperparameters. Architectural fields not shown follow the base UNet configuration for each dataset (`google/ddpm-cifar10-32` or `google/ddpm-celebahq-256` with `sample_size`=128).

Parameter	ColorMNIST	CelebA
Image resolution	32	128
Channels	3	3
Base UNet config	`ddpm-cifar10-32`	`ddpm-celebahq-256` (`sample_size`=128)
Class-embedding rows (incl. null)	121	52
Optimizer	AdamW	AdamW
Learning rate	2×10^{-4}	1×10^{-4}
Weight decay	10^{-6}	10^{-6}
Gradient clip ($\|\nabla\|_2$)	1.0	1.0
Batch size	256	128 (16 per GPU $\times$8)
Training length	100 epochs	500,000 steps
Mixed precision	none	bf16
EMA on UNet weights	no	yes (decay 0.9999)
Null-token dropout p	0.1	0.1
Diffusion schedule	linear β, T=1000	linear β, T=1000
Parameterization	ε-prediction	ε-prediction

F Hyperparameters

F.1 Prototype discovery hyperparameters

The ascent of Eq. (6) starts from the step $\eta = 1-\bar{\alpha}_t$, with which one step is $x \leftarrow x-\sqrt{1-\bar{\alpha}_t}\,\varepsilon_\theta(x,t,\emptyset) = \sqrt{\bar{\alpha}_t}\,\hat{x}_0(x)$. An iterate that jumps across the mode shows as a predicted noise that turns against the predicted noise of the previous step, that is, as a negative inner product of the two. Whenever this happens in an ascent, its η is halved, down to $(1-\bar{\alpha}_t)/16$ at most. No step size is set by hand for any dataset or noise level. Each ascent first runs at its own level t of the grid and then at $t = 50$ (Section C). We model the per-mode covariance $\Sigma_k^{(t_k)}$ as diagonal and estimate its diagonal entries from Hutchinson probes (Hutchinson, 1990) of the Tweedie Jacobian. See Table 7.

Table 7: Prototype discovery settings (per dataset). Both datasets use K=8 prototypes per PoE and the per-coordinate softmax weighting of Eq. (12) with temperature τ. The differences are mostly in the discovery sampling budget, which is larger on ColorMNIST because per-step ascent on a 32×32 tensor is much cheaper than on 128×128. On CelebA an ascent stops once the RMS of the predicted noise is below 0.05.

Hyperparameter	ColorMNIST	CelebA
Number of prototypes K	8	8
Number of PoEs per query	1	4
Selection objective	submodular greedy	submodular greedy
Per-coordinate softmax temperature τ	0.05	0.05
Grid of noise levels (`[start, end, step]`)	[300, 400, 25]	[300, 500, 50]
Refinement level	50	50
Ascents per noise level, from the query	64	8
Ascents per noise level, from the model's own samples	64	32
Ascent steps at each of the two levels	40	at most 60
Ascent step η	$1-\bar{\alpha}_t$, self-halving	$1-\bar{\alpha}_t$, self-halving
Hutchinson probes	4	4
Hutchinson finite-difference step	5×10^{-3}	5×10^{-3}
Unconditional score source	$\emptyset$ token	$\emptyset$ token
Seed	42	42

Table 8: Sampling settings for our method and all baselines.

Hyperparameter	ColorMNIST	CelebA
DDIM scheduler (all methods)	η=0, 50 steps	η=0, 50 steps
Samples per query per method	16	16
Seed (fixed across methods)	42	42
Ours: scale S of the PoE variance	2	1
Ours: block size N	2	4
Composable: guidance strength w	1.0	2.0
Composable: retrieval grid $\mathcal{T}_{\text{eval}}$	$\{50, 100, 150\}$	$\{50, 100, 150\}$
Composable: retrieval temperature τ_{tk}	1.0	1.0
SDEdit: noise level t_0	50 and 200 in Table 1, 50 to 500 in Section H	

F.2 Sampling hyperparameters

G Additional evaluation details

Reference sets. For each OOD class, the reference is the set of held-out members of the same class with the query indices excluded, so no batch is ever compared against its own conditioning image. Every metric is computed per query batch and aggregated to mean ± standard error over the OOD classes.

Why KID rather than FID. At a batch size of 16, FID's plug-in estimator is biased upward by a term that depends on the generator's own covariance, so small-sample FID can reverse orderings between methods of different diversity. KID's U-statistic estimator is unbiased at every sample size (Bińkowski et al., 2018), and we report KID throughout. On CelebA, Inception features separate face identities far more strongly than face classes, so KID there rewards realistic spread more than attribute fidelity and we read it as an image-quality measure only.

KID of the composable diffusion baselines on CelebA. Comp. 1 and Comp. 3 have the lowest KID on CelebA in Table 1, and they reach it at a lower attribute correctness than every other method. Their attribute correctness is 63.4% and 62.2%, where our method reaches 77.0% and held-out members of the query's class, judged in the same way, reach 78.3% (Table 9). The two baselines sample the seen classes that the retrieval returns, so their samples are realistic faces of neighboring seen classes, and, as noted above, KID on CelebA rewards realistic spread more than attribute fidelity. On ColorMNIST, where a seen class differs from the query's class in the digit or in a color, the same baselines have the highest KID of all methods (0.166 and 0.130, against 0.079 for our method).

KID on CelebA can thus be lowered by giving up attribute correctness, and SDEdit and our method show the same relation (Figure 11). For SDEdit, KID falls from 0.218 at 78.3% attribute correctness ($t_0 = 50$) to 0.131 at 73.7% ($t_0 = 500$, Table 10). We therefore compare the KID of two methods only at equal attribute correctness (Section H).

Diversity details. The diversity score of Section 5.3 is the log determinant of a DPP L-ensemble in its quality–diversity form (Kulesza & Taskar, 2012). With relevance to the query $q_i = \exp(-d(x_i, x_q)/h)$ collected in $Q = \operatorname{diag}(q_1, \ldots, q_N)$ and pairwise similarity $S_{ij} = \exp(-d(x_i, x_j)/h)$, a batch of N samples scores

$$\tfrac{1}{N} \log\det(QSQ) \;=\; \tfrac{2}{N} \sum_i \log q_i \;+\; \tfrac{1}{N} \log\det S.$$

Under a DPP with kernel $L = QSQ$, $\det L$ is proportional to the probability of drawing the batch as a set. The factors q_i^2 reward proximity to the query, while $\det S$ rewards diversity within the batch. For the unregularized kernel, the determinant is zero when two samples coincide. We floor the eigenvalues of S at 10^{-12} before computing the log determinant, so coincident samples receive a finite penalty. Higher is better. The distance d is cosine between raw pixel vectors, so no learned encoder enters the judgment, and the bandwidth h is the median nearest-neighbor distance among the class's held-out members, set per class. The score is conditional on the query by design and has

no marginal gold standard. A random batch of 16 real members is fully spread but unanchored, since a marginal draw ignores the query.

Attribute correctness construction. On CelebA, each binary attribute a is judged by a linear rule obtained purely by averaging real images. With μ_a^+ and μ_a^- the mean CLIP ViT-B/32 features of 6,000 labeled training-split images with and without the attribute, a sample carries a when its feature projects beyond the midpoint of μ_a^+ and μ_a^- along $d_a = \mu_a^+ - \mu_a^-$. Three of the 14 attributes (Sideburns, Wearing_Necklace, Bald) are constant across all 74 realized classes and are excluded, leaving 11 informative attributes, so chance is near 50%. On ColorMNIST, every one of the 160 compositions receives a mean image of 200 of its members (training split for seen classes, and for OOD classes held-out members disjoint from both the queries and the evaluation references), and a sample is credited by how many of the three factors of its nearest mean image in pixel space match the target, so chance is 20%. The judging space for each dataset is fixed with real data alone, before any generated image is scored, by two criteria, the ceiling attained on held-out members and the stability of that ceiling under small perturbations of real members. Pixel space passes both on ColorMNIST, where CLIP is unstable at 32×32. CLIP passes both on CelebA, where pixel distance is dominated by pose and illumination. No judge involves training.

Threshold variant. Table 9 compares the midpoint threshold of the main text with a calibrated variant whose per-attribute threshold maximizes accuracy on the labeled real images, accounting for attribute imbalance. The composable diffusion baselines shift upward by about three points, the other methods change by at most 0.3 points, and the order of our method and the baselines does not change.

Table 9: CelebA attribute correctness (%, mean$_{\pm SE}$) under the two threshold rules.

Method	Midpoint	Calibrated
Comp. 1	$63.4_{\pm 1.2}$	$66.0_{\pm 1.2}$
Comp. 3	$62.2_{\pm 1.0}$	$66.0_{\pm 1.4}$
SDEdit (t_0=50)	$78.3_{\pm 2.1}$	$78.6_{\pm 1.1}$
SDEdit (t_0=200)	$77.1_{\pm 2.1}$	$77.4_{\pm 1.1}$
Ours	$77.0_{\pm 2.0}$	$76.8_{\pm 1.1}$
Real (held-out)	$78.3_{\pm 2.0}$	$78.8_{\pm 1.0}$

H SDEdit over the noise level

SDEdit has one hyperparameter, the noise level t_0 to which the query is noised before the unconditional reverse chain. We ask whether the comparison of Table 1 depends on it, and sweep t_0 on both datasets, on the same queries and with the same 16 samples per query as every other method.

Table 10: SDEdit over the noise level t_0, and our method, on the full protocol of Section 5. Attribute correctness (%), KID per query batch, and diversity, averaged over the held-out classes.

	ColorMNIST			CelebA		
Method	Attr. Corr. ↑	KID ↓	Diversity ↑	Attr. Corr. ↑	KID ↓	Diversity ↑
SDEdit, t_0 = 50	92.9	0.107	−3.40	78.3	0.218	−4.21
SDEdit, t_0 = 100	92.8	0.094	−2.62	78.1	0.197	−3.57
SDEdit, t_0 = 150	92.7	0.089	−2.37	77.7	0.189	−3.39
SDEdit, t_0 = 200	92.3	0.082	−2.17	77.1	0.179	−3.21
SDEdit, t_0 = 250	91.8	0.078	−2.12	76.7	0.174	−3.10
SDEdit, t_0 = 300	89.9	0.070	−2.28	76.0	0.165	−2.96
SDEdit, t_0 = 400	78.5	0.060	−4.17	74.8	0.150	−2.69
SDEdit, t_0 = 500	59.2	0.054	−8.77	73.7	0.131	−2.38
Ours	92.5	0.079	−2.12	77.0	0.166	−3.13

Table 10 shows that a larger t_0 lowers the KID of SDEdit only together with its attribute correctness, and that at equal attribute correctness our method has the lower KID and the higher diversity on both datasets. On CelebA, SDEdit at $t_0 = 200$ matches our attribute correctness (77.1% vs. 77.0%). There our KID is lower in all 23 held-out classes, with a paired difference of $-0.013_{\pm 0.002}$ (mean and standard error over classes), and our diversity is higher in 21 of them. On ColorMNIST, our attribute correctness (92.5%) lies between SDEdit at $t_0 = 150$ and $t_0 = 200$, and our KID (0.079) is lower than both (0.089 and 0.082).

Figure 11 plots attribute correctness against KID and against diversity for both methods. For our method the points follow the scale S of the PoE variance (Section C), which plays the role that t_0 plays for SDEdit, since a wider PoE lowers KID and attribute correctness together. At every attribute correctness that both methods reach, our method has the lower KID.

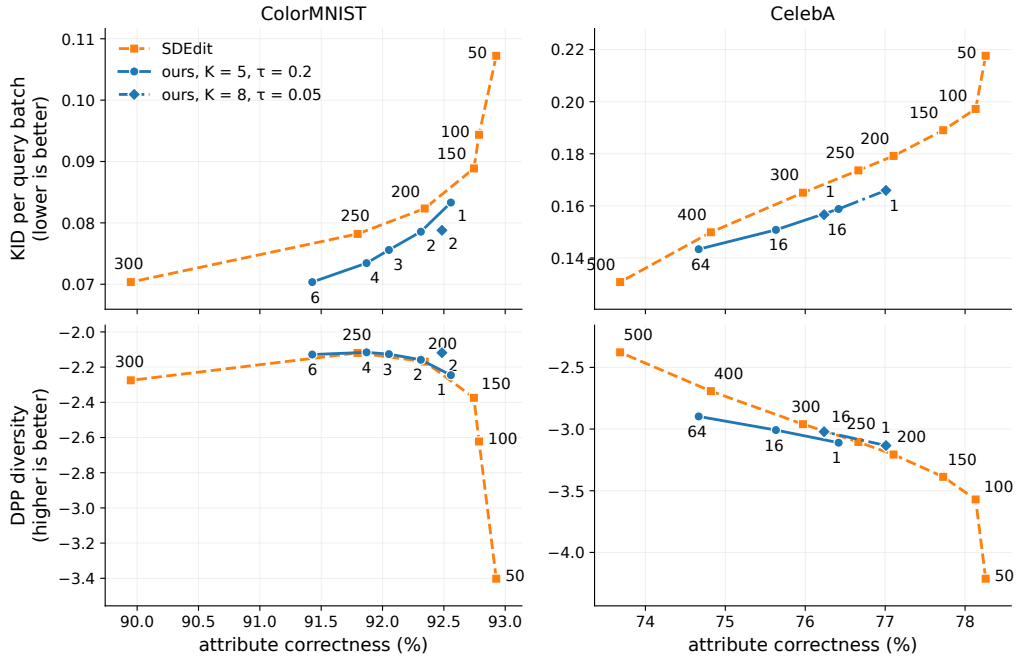


Figure 11: Attribute correctness against KID per query batch (top) and against diversity (bottom) on ColorMNIST and CelebA, for SDEdit over t_0 and for our method over the scale S of the PoE variance, with $K = 5$, $\tau = 0.2$ and with $K = 8$, $\tau = 0.05$. The numbers next to the points are t_0 and S.

I Number of prototypes and prototypes from the model's own samples

Number of prototypes. We ask how many prototypes the composition needs, and vary the number of selected prototypes K with all other settings as in Table 7. Table 11 and Figure 12 show that attribute correctness rises with K and saturates between $K = 5$ and $K = 8$ on both datasets, from 65.7% at $K = 1$ to 92.5% at $K = 8$ on ColorMNIST and from 72.8% to 77.0% on CelebA.

Prototypes taken only from the model's own samples. We ask whether the samples are built from the query. In the second pool of Table 11 no ascent starts at the query. Every candidate is reached from a noised copy of one of the model's own unconditional samples, 128 per noise level, and the query enters only through the selection and the weights.

Table 11 shows that the result does not depend on ascents from the query. With $K = 8$ and no ascent from the query, the method reaches 92.5% attribute correctness on ColorMNIST and 77.1% on CelebA, against 92.5% and 77.0% with ascents from the query, and KID and diversity are unchanged. Ascents from the query help only when few prototypes are selected (65.7% against 51.2% at $K = 1$ on ColorMNIST).

Table 11: Attribute correctness (%) against the number of selected prototypes K, for the pool of Table 7 and for a pool with no ascent from the query, where every candidate is reached from one of the model's own unconditional samples. "Carried" is the share of queries (%) whose K prototypes, in the second pool, jointly carry every factor of the query's class, the three factors on ColorMNIST and the 11 attributes on CelebA.

	K	1	2	3	5	8	12
ColorMNIST	Pool with ascents from the query	65.7	90.2	91.9	92.4	92.5	92.5
	No ascent from the query	51.2	88.7	91.4	92.3	92.5	92.5
	Carried	2.7	48.2	69.2	84.2	90.5	95.0
CelebA	Pool with ascents from the query	72.8	74.3	75.5	76.5	77.0	77.2
	No ascent from the query	59.6	69.5	74.2	76.3	77.1	77.3
	Carried	2.5	22.7	45.0	73.3	91.7	97.7

The unseen class is composed from prototypes of seen classes. On ColorMNIST we read the class of every selected prototype with the judge of Section G, and at least 95% of the prototypes are seen classes. A single prototype carries 1.5 of the three factors of the query on average, and eight prototypes carry 2.9. Accordingly, the K prototypes jointly carry every factor of the query for 2.7% of the queries at $K = 1$ and for 90.5% at $K = 8$ (Table 11).

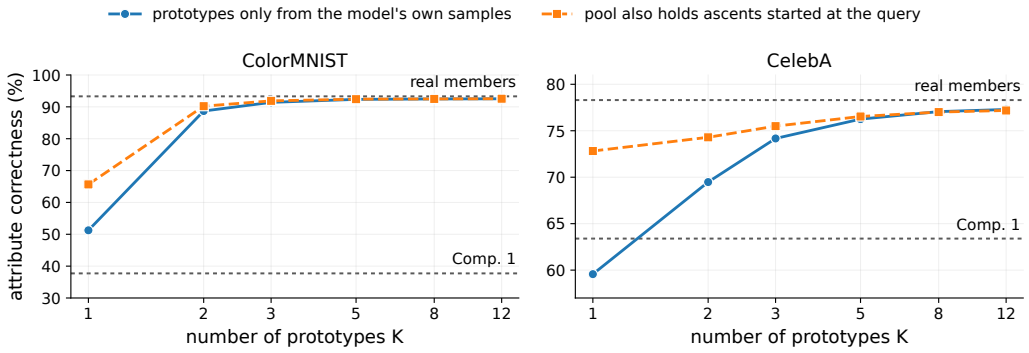


Figure 12: Attribute correctness against the number of selected prototypes K on ColorMNIST and CelebA, for prototypes taken only from the model's own samples and for the pool that also holds ascents from the query. The dashed lines mark held-out real members of the class and Comp. 1.

Temperature. On ColorMNIST with $K = 5$, attribute correctness rises from 87.7% at the temperature $\tau = 1.0$ of Eq. (12) to 92.3% at $\tau = 0.2$ and 92.4% at $\tau = 0.05$. A high temperature averages prototypes that disagree in a dimension, for example two background colors into a color in between, and a low temperature lets the prototype that explains the query best in that dimension decide.

Several PoEs per query. Figure 13 shows the four PoEs of two CelebA queries (Section C), with the prototypes, the mean, and the samples of each.

J Compute

Backbone training takes roughly 1.5 hours on a single NVIDIA H200 GPU for ColorMNIST and roughly 7 hours on eight NVIDIA H200 GPUs for CelebA. Test-time prototype discovery and PoE sampling both run on a single GPU. The full pipeline of Section 3 (mode ascent, Hutchinson estimation of the local Hessian, submodular selection, PoE composition, and sampling) takes under 15 seconds per query.



Figure 13: The four PoEs of two held-out CelebA queries. Each row shows one PoE with its $K = 8$ selected prototypes, its mean, and its four samples.

K Visualizations

We present additional visualizations for both datasets in Figure 14.

L Generalization to unseen primitives

Prototype discovery operates on the score field induced by the trained DDPM, so the prototypes it can surface are confined to regions where that score field is informative. Figure 15 illustrates this boundary with two queries that share an in-palette pink digit but differ in background. The first places the digit on a black background, which lies inside the convex hull of the four trained backgrounds (*deep red*, *navy*, *dark purple*, *dark brown*), all of which are dark and unsaturated. The second places the digit on a white background, which falls far outside that hull. The discovered prototypes and the PoE samples reproduce the held-out black background as a coherent novel primitive, while the white query collapses onto trained dark backgrounds with the pink digit preserved but the brightness of the background lost. The discovery procedure therefore inherits the inductive biases of the backbone. It can compose new primitives that interpolate between seen ones, but it cannot fabricate primitives that extrapolate beyond the training support.

M Broader Impacts

A potential positive impact is reducing the data and compute needed to generate novel visual compositions at test time. A potential negative impact is that the same capability could lower the barrier to generating face images with novel attribute combinations, which could be misused for misleading imagery. However, the method operates on small-scale class-conditional DDPMs and does not approach the fidelity of modern text-to-image systems.

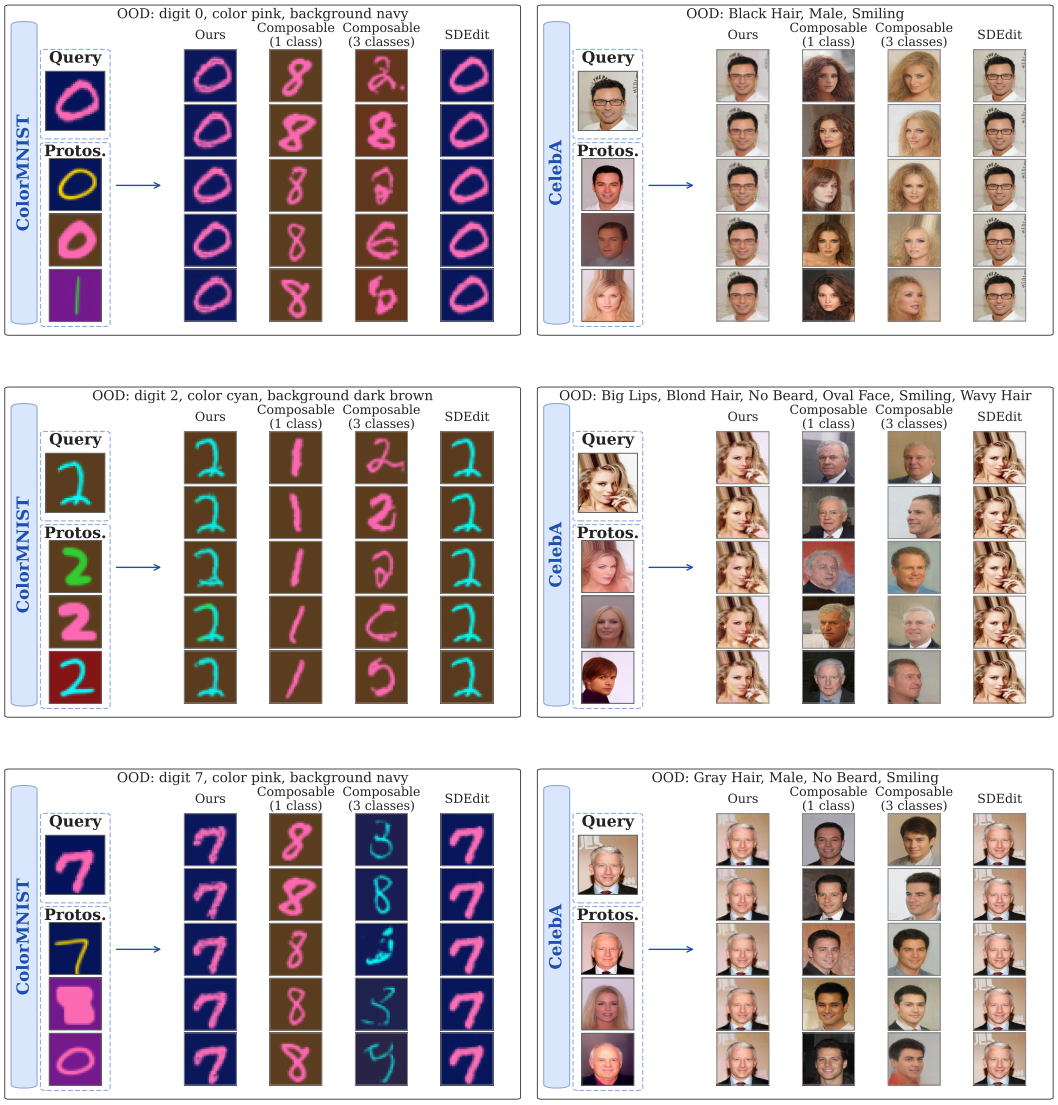


Figure 14: Additional qualitative results for ColorMNIST and CelebA, for three more held-out queries per dataset in the layout of Figure 3. On CelebA the prototypes are those of the first of the four PoEs, and the samples of our method are one from each PoE and a second from the first.

N Licenses for Existing Assets

Table 12 lists the licenses of the existing assets that we use.

Table 12: Licenses for existing assets.

Asset	License	Source
MNIST	CC BY-SA 3.0	MNIST website
CelebA	Non-commercial research	CelebA website
HF diffusers	Apache 2.0	GitHub
DiT-XL/2	CC BY-NC 4.0	Hugging Face
Latent diffusion text-to-image model	Apache 2.0	Hugging Face

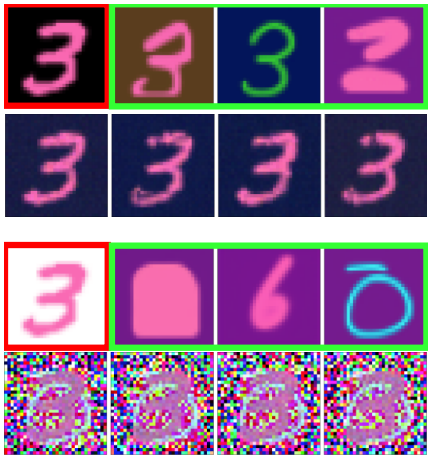


Figure 15: Prototype discovery for novel ColorMNIST background primitives with a fixed pink digit. Each panel contains the query (red), three discovered prototypes (green), and four PoE samples. **Top:** interpolation to a held-out black background produces coherent prototypes and samples. **Bottom:** extrapolation to a white background fails: prototypes remain near trained dark backgrounds and PoE samples collapse to salt-and-pepper texture.